

Traffic Light

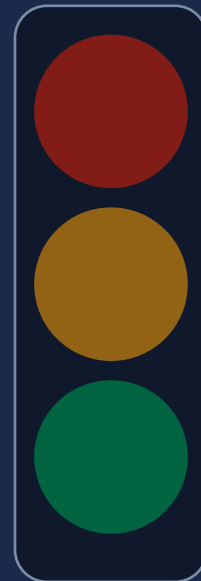
Decision Controller

EEE120 Digital Design Fundamentals — Final Project

Group 4 | Usmon Shamirzayev & Team

Course: EEE120 | AUT Tashkent | May 2026

GitHub Repository: https://github.com/UsmonMD/EEE120_Final_Project_Group_4



Team Members

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Problem Statement

The Challenge

Small road crossings lack intelligent signal control. Without logic-based systems:

- Emergency vehicles get delayed
- Pedestrians have no safe crossing phase
- Side roads are ignored when main is empty
- Fixed timers waste time when roads are clear

Our Solution

Design a combinational logic circuit that automatically decides which signal turns green based on real-time road conditions — no clock, no memory, pure logic.



Real-World Relevance

Smart Cities

Autonomous Vehicles

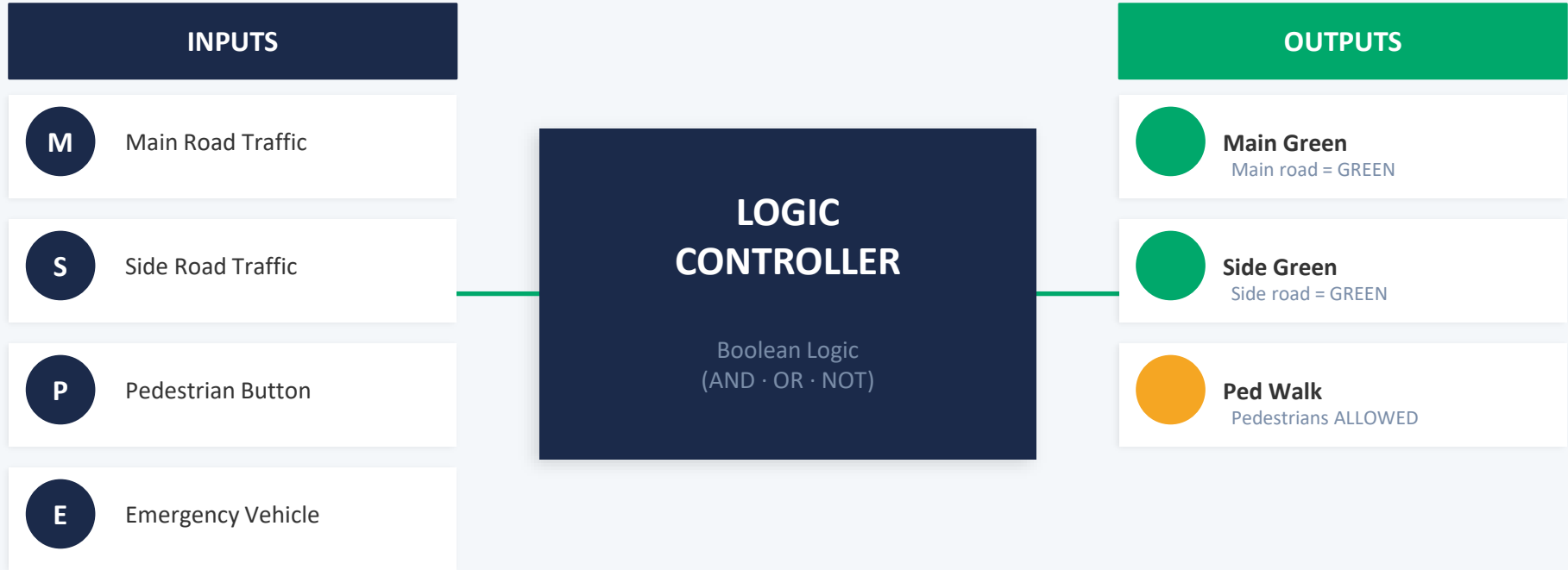
IoT Traffic

Urban Infrastructure

Embedded Systems

Road Safety

System Overview



Logic Design — Boolean Expressions

1 Emergency (E=1)

Overrides ALL.
Intersection cleared.
Everything is RED.

2 Pedestrian (P=1)

Ped Walk ON.
Both roads RED.

3 Side Only (S=1,M=0)

Side road GREEN.
Main stays RED.

4 Default

Main road GREEN.
(default state)

Boolean Expressions

$$\text{Main_Green} = E + (E' \cdot P' \cdot (M + S'))$$

$$E' \cdot P$$

$$\text{Side_Green} = E' \cdot M' \cdot P' \cdot S$$

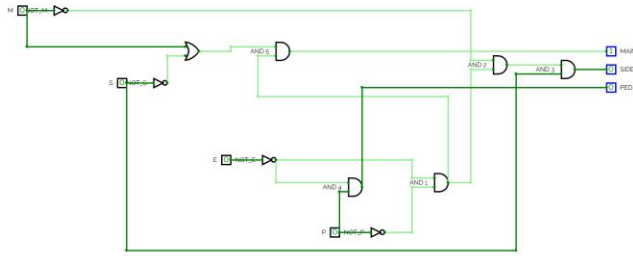
$$\text{Ped_Walk} =$$

' = NOT · = AND + = OR | Gates: 3× NOT + 4× AND + 4× OR = 10 total ✓

Truth Table

M	S	P	E	Main	Side	Ped	State / Reason
0	0	0	0	1	0	0	Default → Main Green
0	1	0	0	0	1	0	Side only → Side Green
0	0	1	0	0	0	1	Ped pressed → Walk
1	0	0	0	1	0	0	Main traffic → Main Green
1	1	0	0	1	0	0	Both traffic → Main priority
1	1	1	0	0	0	1	Ped overrides traffic
0	0	0	1	0	0	0	Emergency → All RED
1	1	1	1	0	0	0	Emergency overrides all

Showing 8 key rows of 16 total combinations. Full table available in README.



Circuit Components

- **INPUT (x4)**
M, S, P, E
- **NOT (x4)**
E', M', P'
- **AND (x5)**
Combine signals
- **OR (x1)**
Final outputs
- **OUTPUT (x3)**
Main, Side, Ped

Total: 10 gates ✓
(min. requirement: 5)

Python Software Prototype

```
# Boolean logic – matches circuit exactly
def traffic_logic(M, S, P, E):
    NOT_E = 1 - E
    NOT_M = 1 - M
    NOT_P = 1 - P
    NOT_S = 1 - S
    main_green = E | (NOT_E & NOT_P
                     & (M | NOT_S))
    side_green = NOT_E & NOT_M & NOT_P & S
    ped_walk = NOT_E & P
    return main_green, side_green, ped_walk
```



Manual input mode

4 road conditions



Live signal display

Emoji output



Python 3

No external libs



Auto truth table

All 16 combos



Explains reasoning

For each decision

```
PS D:\EEE120_Final_Project_Group4> & "d:\Development\python\python.exe"
=====
TRAFFIC LIGHT DECISION CONTROLLER
EEE120 Final Project - Group 4
=====

MENU:
1. Enter road conditions manually
2. Run automated truth table test
3. Exit

Choose (1/2/3): 1

Enter road conditions (1-Yes, 0-No):
Main road has traffic? : 1
Side road has traffic? : 1
Pedestrian button pressed? : 0
Emergency vehicle detected? : 0

=====
TRAFFIC SIGNAL DECISION
=====
Inputs:
M (Main road traffic) : YES
S (Side road traffic) : YES
P (Pedestrian button) : NO
E (Emergency vehicle) : NO

Outputs:
Main Road Signal      : 🟢 GREEN
Side Road Signal      : 🟡 RED
Pedestrian Crossing    : 🚫 NOT ALLOWED

🟢 MAIN ROAD PHASE: Normal traffic flow.

=====

MENU:
1. Enter road conditions manually
2. Run automated truth table test
3. Exit

Choose (1/2/3): 2

=====
TRUTH TABLE - All 16 Input Combinations
=====
M  S  P  E | Main Side Ped
-----
0  0  0  0 | 1  0  0
0  0  0  1 | 0  0  1
0  0  1  0 | 0  1  0
0  0  1  1 | 0  0  1
0  1  0  0 | 1  0  0
0  1  0  1 | 1  0  0
0  1  1  0 | 1  0  1
0  1  1  1 | 1  0  1
1  0  0  0 | 1  0  0
1  0  0  1 | 1  0  0
1  0  1  0 | 1  0  0
1  0  1  1 | 1  0  0
1  1  0  0 | 1  0  0
1  1  0  1 | 1  0  0
1  1  1  0 | 1  0  0
1  1  1  1 | 1  0  0

=====

MENU:
1. Enter road conditions manually
2. Run automated truth table test
3. Exit

Choose (1/2/3):
```

AI / LLM Usage Reflection

 We used Claude (Anthropic) as an AI assistant — AI was NOT used to fully make the project

Task	What AI helped with	What we understood & verified
Output formatting	Improve readability and clarity of program output	We adjusted output to clearly display signal states and meanings
Code comments & explanation	Generated structured comments explaining each code part	We reviewed all comments and verified they correctly describe the logic
Debugging & problem solving	Assisted with Python syntax and logic errors	We tested each fix and verified correctness through multiple test cases
Presentation preparation	Helped structure and format the presentation slides	We reviewed, edited, and ensured the presentation reflects our understanding

Conclusion & Future Work



What Worked

- Boolean logic correctly controls all 3 outputs
- Python matches circuit logic exactly
- Emergency override works in all test cases
- Clean CircuitVerse simulation with labels



Challenges

- Setting correct gate priorities
- Handling conflicting inputs (P=1 and E=1)
- Mapping Python operators to logic gates
- Keeping circuit readable in CircuitVerse



Future Improvements

- Sequential logic with flip-flops for timed phases
- LCD countdown timer display
- 4-way intersection support with more inputs
- Night mode — flashing yellow
- Arduino/Raspberry Pi physical implementation

Thank You! — EEE120 Group 4 — Traffic Light Decision Controller